

Notes: Pycia and Woodward (JPE, 2026)

A Case for Pay-as-Bid Auctions

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1 Big picture

- **Question.** What does the equilibrium of a pay-as-bid multiunit auction look like, how should a revenue-maximizing seller design such an auction, and how does pay-as-bid compare with uniform-price auctions?
- **Setting.** The paper studies multiunit auctions for homogeneous goods such as treasury securities and electricity. Bidders are symmetrically informed about a common signal s , while the seller may know less about bidders' information when designing the auction.
- **Core challenge.** Pay-as-bid auctions are widely used in practice, but equilibrium behavior is much less understood than in uniform-price auctions. This leaves open basic design questions: what reserve price and quantity distribution are optimal, how much should the seller reveal about supply, and why do empirical studies often find rough revenue equivalence across formats?
- **Main conceptual contribution.** The paper turns pay-as-bid from a format with limited tractable theory into one with a sharp equilibrium structure: minimum clearing price, uniqueness of pure-strategy equilibrium, and a closed-form bid representation.
- **Main equilibrium result.** In the symmetric-information environment, pure-strategy equilibrium is essentially unique. The equilibrium bid for any unit is a weighted average of marginal values on that and higher units.
- **Main design result 1.** Optimal pay-as-bid auctions are transparent: revenue is maximized by deterministic supply rather than random supply.
- **Main design result 2.** Full disclosure of realized supply is optimal. If the seller has information about supply, the seller wants to reveal it before bidding.
- **Main comparison result.** In the seller's design game, pay-as-bid raises weakly more expected revenue than any perfect Bayesian equilibrium of the uniform-price design game.
- **Why this is not the usual “pay-as-bid always beats uniform-price” claim.** The result is about *optimal design*. The paper does not say that every arbitrary pay-as-bid auction dominates every arbitrary uniform-price auction. Rather, once reserve price, supply, and disclosure are chosen optimally, pay-as-bid is weakly revenue superior.
- **Why approximate revenue equivalence still appears in the data.** The paper shows that revenue equivalence can emerge when the seller would choose uniform price only in seller-optimal equilibria that match pay-as-bid revenue, or when supply randomness is so small that bids in both formats become nearly flat.
- **Welfare message.** Welfare comparisons are not one-sided. Pay-as-bid may yield higher welfare in some environments, but welfare is ambiguous in general because it depends on the environment and equilibrium selection in uniform price.
- **Economic takeaway.** The paper makes a case for pay-as-bid auctions by showing that they combine equilibrium uniqueness, transparent design, and weak revenue dominance over uniform-price auctions in the relevant design problem.

2 Model environment and key objects

2.1 Economic setting and why the paper focuses on it

There are $n \geq 2$ bidders. Bidder i 's marginal valuation for quantity q is

$$v(q; s), \tag{1}$$

where s is a common signal observed by all bidders but not necessarily by the seller at the design stage. For any fixed s , the paper writes the marginal value simply as $v(q)$.

Interpretation. The paper is built for environments such as treasury auctions in which bidders can often infer the asset's value from surrounding market prices, so bidder asymmetries are limited even if the seller must commit to a design before all relevant information is known.

2.2 Shape restrictions on values

For each signal realization, $v(\cdot; s)$ is:

- nonnegative,
- strictly decreasing wherever it is strictly positive,
- Lipschitz continuous,
- and almost everywhere differentiable.

The paper also assumes values fall fast enough that the monopoly problem

$$\max_{q \geq 0} q v(q; s) \tag{2}$$

has a finite solution.

Interpretation. This keeps demand well behaved and ensures the seller's optimal quantity problem is meaningful.

2.3 Supply, reserve price, and effective quantity

Aggregate supply Q is drawn from a distribution F with support $[0, \bar{Q}]$. The seller also sets a reserve price $R \geq 0$.

Because the reserve can bind, the paper defines the *effective quantity* actually allocated:

$$Q_R(Q, s) = \begin{cases} Q, & \text{if } R = 0, \\ \min\{Q, n v^{-1}(R; s)\}, & \text{if } R \in (0, v(0; s)). \end{cases} \tag{3}$$

The corresponding effective maximum quantity is

$$\bar{Q}_R = Q_R(\bar{Q}, s). \tag{4}$$

Interpretation. If the reserve price is high enough, the seller may refuse to allocate some of the available supply. So the relevant “top quantity” for equilibrium analysis is not always the physical maximum $\bar{Q}$ but the effective maximum $\bar{Q}_R$.

2.4 Bid functions and the stop-out price

Each bidder submits a weakly decreasing bid function

$$b_i(q) : [0, \bar{Q}] \rightarrow \mathbb{R}_+. \quad (5)$$

The stop-out price is the highest price consistent with total demand still covering realized supply:

$$p^* = \max \left\{ R, \sup \left\{ p' : q_1 + \dots + q_n \geq Q \text{ for all } q_1, \dots, q_n \text{ such that } b_1(q_1), \dots, b_n(q_n) \leq p' \right\} \right\}. \quad (6)$$

Bidder i then receives

$$q_i = \max \{ q' : b_i(q') \geq p^* \}, \quad (7)$$

with pro-rata rationing on the margin if necessary.

Interpretation. In pay-as-bid, each unit won is paid at the bidder's own bid. This means a bidder must think not only about the marginal unit that sets the clearing price, but also about the inframarginal units on which the bid is paid whenever it lies above the clearing price.

2.5 Bidder payoff

Utility is quasi-linear:

$$u_i(b_i) = \int_0^{q_i(p^*)} (v(x) - b_i(x)) dx. \quad (8)$$

Interpretation. This is the source of bid shading. Raising a bid can help win an additional unit, but if the bid is high enough it also raises payment on units that would already have been won.

2.6 Seller's design problem

The seller chooses:

- the reserve price R ,
- the distribution of supply (or, in the appendix, elastic supply curves),
- and potentially a disclosure policy mapping realized supply into public messages.

The paper's main design results are stated for a seller who maximizes expected revenue and typically must choose the design before observing the bidders' signal s .

Interpretation. The paper separates *equilibrium analysis* from *design analysis*. First it solves how bidders behave for a given supply distribution and reserve. Then it asks what the seller should optimally choose.

3 Models and theoretical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper's logic has four moving parts:

- bidders respond to *residual supply*, not to a fixed quantity known with certainty;
- the more uncertain supply is, the more bidders hedge by bidding relatively aggressively on low quantities;
- in pay-as-bid, unlike in uniform price, pure-strategy equilibrium is unique, so design can be studied without a large equilibrium-selection problem;
- random supply and opacity hurt the seller because they flatten the information structure and soften bids less efficiently than a transparent deterministic design.

Interpretation. A lot of the paper can be summarized in one sentence: once you know the minimum clearing price and the shape of residual supply, the whole pay-as-bid equilibrium becomes tractable enough to solve the seller's design problem.

3.2 Theorem 1: minimum clearing price

The key starting point is the minimum clearing price theorem:

$$p(\bar{Q}_R) = v\left(\frac{\bar{Q}_R}{n}\right). \quad (9)$$

Interpretation. At the effective maximum quantity, the clearing price must equal each bidder's marginal value on the last unit they could receive. If a bidder were strictly below value there, they could profitably shade upward a little, win more quantity, and pay only slightly more. This theorem pins down the lower boundary condition for equilibrium bidding.

3.3 Theorem 2: essential uniqueness

Conditional on existence, the Bayesian Nash equilibrium is essentially unique.

Interpretation. "Essentially unique" means all equilibria agree on the bids for quantities that can be won with positive probability. So clearing prices, allocations, and payments are pinned down on the equilibrium path. This is a sharp contrast with uniform-price auctions, where multiplicity is pervasive.

3.4 Theorem 3: bid representation

For any quantity $q \in [0, \bar{Q}_R/n]$, define the weighting distribution

$$F_{Q,n}(x) = 1 - \left(\frac{1 - F(x)}{1 - F(Q)}\right)^{(n-1)/n}, \quad x \in [Q, \bar{Q}]. \quad (10)$$

Then the essentially unique equilibrium bid of each bidder is

$$b_i(q) = \int_{nq}^{\bar{Q}} v\left(\frac{\min\{x, \bar{Q}_R\}}{n}\right) dF_{nq,n}(x). \quad (1)$$

The associated clearing price for realized supply Q is

$$p(Q) = b_i\left(\frac{Q}{n}\right) = \int_Q^{\bar{Q}} v\left(\frac{\min\{x, \bar{Q}_R\}}{n}\right) dF_{Q,n}(x). \quad (2)$$

Interpretation. The bid on quantity q is a weighted average of marginal values not only at q but also at higher quantities. The weighting depends only on the supply distribution and the number of bidders. This is the paper's main tractability result.

What the bid formula means economically

The weighting distribution places more mass on lower quantities when there are more bidders, because competition forces bidders to retain less rent on marginal units. In the large- n limit, the bid becomes the expected market-clearing value conditional on the bidder receiving at least that unit.

Interpretation. The formula says pay-as-bid bidding is neither arbitrary nor purely local. A bidder computes how a unit today is related to future higher-quantity states in which the same bid may also matter.

Special case: deterministic supply

If supply is deterministic, the bid function is flat up to $\bar{Q}_R/n$.

Interpretation. This is one of the most important empirical implications of the paper. When quantity is nearly known, there is little reason to make bids slope down. Bidders know what they will win, so shading becomes nearly uniform across relevant quantities.

3.5 Theorem 4: existence

The paper defines

$$U(\hat{q}; q) = (v(q) - b(\hat{q})) [1 - F(q + (n-1)\hat{q})]. \quad (11)$$

This is the contribution of unit q to expected utility when the bidder submits the bid intended for quantity $\hat{q}$.

Theorem 4 states that a pure-strategy Bayesian Nash equilibrium exists whenever, for almost every $q \in [0, \bar{Q}/n]$, the derivative of $U(\cdot; q)$ is zero only at global maxima.

Interpretation. The paper does not simply assume equilibrium exists. It proves a sufficient condition that says the candidate bid from theorem 3 is genuinely pointwise optimal.

Corollary 1: existence with many bidders

If marginal values are smooth enough and per-capita supply has a density that is bounded away from zero with bounded derivative, then equilibrium exists once the number of bidders is sufficiently large.

Interpretation. The equilibrium is not a knife-edge object. In regular environments it exists automatically in sufficiently competitive markets.

3.6 Comparative statics: flat bids, more bidders, and mergers

The paper then draws several implications from the bid formula.

Corollary 2: concentrated supply implies flat bids

If supply is tightly concentrated around quantity Q^* , then equilibrium bids for quantities below Q^*/n become arbitrarily close to $v(Q^*/n)$.

Interpretation. As supply uncertainty vanishes, pay-as-bid bids become approximately flat. This is the formal bridge from the theory to observed near-flat bids in some treasury auctions.

Corollary 3: more bidders increase bids and revenue

With more bidders:

- bids are higher,
- seller revenue is higher,
- and each bidder's profit is lower,

both when total supply is fixed and when per-capita supply is fixed.

Interpretation. Competition compresses bid shading. The paper also notes that merging two markets with the same per-capita supply distribution raises revenue because the merged market has more competition.

3.7 Theorem 5: transparency of optimal supply

The seller's revenue under nondeterministic supply is strictly lower than under optimal deterministic supply. The optimal deterministic quantity is the solution to the monopoly problem when demand is uncertain.

A central bound behind the result is that, for any supply distribution F ,

$$\mathbb{E}_{s,Q}[p_F(Q; s)] \leq \int_0^{\bar{Q}_R} \mathbb{E}_s[p_{dQ}(Q; s)] dF(Q), \quad (3)$$

where $p_F(Q; s)$ is revenue when bidders face the supply distribution F and $p_{dQ}(Q; s)$ is revenue when supply is deterministically fixed at Q .

Interpretation. Randomizing supply cannot beat the envelope of deterministic monopoly-style quantities. The seller does not gain by creating quantity uncertainty; instead, the seller should pick one optimal quantity cap and one reserve price.

Why the transparency result is nontrivial

A random supply distribution could in principle look attractive because sometimes the seller sells less than the monopoly quantity, while in states with higher quantity bidders may still pay relatively high bids. The theorem shows this intuition is false once equilibrium bids are taken into account.

Interpretation. This is one of the paper’s most surprising results: pay-as-bid’s optimal design is simpler than one might expect. The best auction is transparent, not opaque.

3.8 Proposition 1: no need for extra randomness in supply caps

If the seller’s underlying supply is itself random and the seller can only cap realized supply from above, then introducing additional randomness in the cap cannot help. The optimal cap is deterministic.

Interpretation. Even when the auctioneer faces exogenous quantity uncertainty, the right response is to remove extra randomness where possible, not to add more.

3.9 Theorem 6: optimality of full disclosure

Suppose the seller can commit to a disclosure policy that maps realized supply into a public message before bids are submitted. Then expected revenue is maximized by fully revealing realized supply.

Interpretation. If the seller knows more about current supply than bidders do, the seller wants to tell them. Hiding supply information makes bidders hedge and generally softens the bids in the wrong way.

Why the disclosure theorem follows from the earlier analysis

Fix a disclosure strategy. Any message induces a posterior distribution of supply $\hat{F}$. Revenue after that message is bounded above by the expected revenue the seller would get if the actual realization were revealed. So state by state, disclosure dominates information pooling.

Interpretation. Theorem 6 is the information-design counterpart of theorem 5. Deterministic supply is best ex ante, and truthful revelation of realized supply is best ex post.

3.10 Uniform-price benchmark and lemma 1

The paper then turns to the uniform-price design game. Unlike pay-as-bid, the uniform-price auction has many equilibria, and this means seller-optimal design is intertwined with equilibrium selection.

Lemma 1 shows that, under a rich enough value space, there is a perfect Bayesian equilibrium of the uniform-price design game in which the seller chooses reserve and supply arbitrarily close to the pay-as-bid optimum. The logic uses:

- a *semitruthful* equilibrium as the on-path continuation equilibrium,
- and a low-revenue *robust* equilibrium as an off-path punishment.

Interpretation. This is why the seller’s problem looks much messier in uniform price. Equilibrium multiplicity breaks the transparent-design logic that works so cleanly in pay-as-bid.

3.11 Theorem 7: revenue comparison with uniform-price auctions

The main comparison theorem states:

$$\mathbb{E}[\text{revenue in the pay-as-bid design game}] \geq \mathbb{E}[\text{revenue in any perfect Bayesian equilibrium of the uniform-price design game}] \quad (12)$$

Interpretation. Once the seller chooses the auction design optimally, pay-as-bid is weakly revenue superior. This is not because pay-as-bid always has steeper bids at a given arbitrary quantity schedule, but because the whole design problem is better behaved.

When is the inequality strict?

The inequality is strict for generic non-semitruthful equilibria of the uniform-price auction. The only way uniform price can match pay-as-bid is through seller-optimal equilibria in which bidders are semitruthful at the margin.

Interpretation. Revenue equivalence is therefore a knife-edge benchmark inside the uniform-price world, not the generic outcome.

Corollaries 5 and 6

Two especially important corollaries follow.

- **Corollary 5.** In a design game where the seller chooses the format itself, all perfect Bayesian equilibria are revenue equivalent because the seller chooses uniform price only if it matches pay-as-bid revenue.
- **Corollary 6.** If the seller knows bidders' signal s when designing the auction, the optimally designed uniform-price auction has a unique equilibrium and is revenue equivalent to pay-as-bid.

Interpretation. These corollaries explain why empirical work can find approximate revenue equivalence even though the generic pay-as-bid design result is revenue dominance.

3.12 Example: closed-form optimal pay-as-bid design

In the paper's worked example, marginal values are linear:

$$v(q; s) = s - \rho q, \quad (13)$$

and the common signal is uniformly distributed on $[\underline{s}, \bar{s}]$. Then the optimal pay-as-bid auction has

$$Q^{*,\text{PAB}} = \left(\frac{3\bar{s} + \underline{s}}{8\rho} \right) n, \quad R^{*,\text{PAB}} = \frac{\bar{s} + 3\underline{s}}{8}. \quad (14)$$

Interpretation. This example shows how the design problem collapses to a monopoly calculation in pay-as-bid, while the optimal uniform-price design remains tangled up with equilibrium selection.

4 Identification and comparison logic (what makes the results credible)

4.1 What pins down the pay-as-bid equilibrium

Theorems 1 to 3 fit together tightly:

- theorem 1 fixes the boundary condition at the effective maximum quantity;
- theorem 2 rules out multiple pure-strategy equilibria on relevant quantities;
- theorem 3 turns the equilibrium into an explicit bid formula.

Interpretation. The paper’s credibility comes from this sequence. It does not merely guess a plausible bidding rule. It first pins down the minimum clearing price, then uses that boundary condition to solve the equilibrium globally.

4.2 Why uniqueness matters so much

Uniqueness is not just aesthetically convenient. It matters for:

- counterfactual design,
- empirical interpretation,
- and comparison with uniform-price auctions.

Interpretation. If a format has many equilibria, the seller’s design problem is partly an equilibrium-selection problem. Pay-as-bid largely escapes that issue; uniform price does not.

4.3 Why theorem 5 is more than a monopoly analogy

It would be easy to misread theorem 5 as saying “the seller is just a monopolist.” That is not the logic. The key point is that bidders’ equilibrium bids against a random supply distribution can be upper bounded by a weighted average of deterministic-supply monopoly revenues. Only after proving that bound does the monopoly analogy emerge.

Interpretation. The seller’s optimal design is not assumed to be deterministic because monopoly pricing is familiar; it is derived from the equilibrium structure of pay-as-bid bidding.

4.4 Why theorem 6 is a strong information-design statement

The full-disclosure result is strong for two reasons:

- it does not require the seller to reoptimize the reserve or supply cap after disclosure;
- and it holds even more forcefully if the seller cannot commit, because proving optimality under commitment automatically implies no-commitment disclosure is not worse.

Interpretation. The paper says opacity is not a feature of the optimal pay-as-bid auction. If the seller knows the supply realization, the seller should tell bidders.

4.5 Why the revenue comparison is a design result, not a fixed-auction result

A lot of older debate asks whether pay-as-bid or uniform price raises more revenue *holding the auction environment fixed*. This paper instead compares the two *design games*. The seller chooses reserve, supply, and (implicitly) relies on an equilibrium selection in uniform price.

Interpretation. This is why the theorem can say pay-as-bid weakly dominates, while the empirical literature can still show near-equivalence. The seller chooses uniform price only when it happens to be as good as pay-as-bid.

4.6 What the paper can and cannot distinguish

The paper is strongest on:

- the structure of pay-as-bid equilibrium under symmetric bidder information,
- transparent optimal design,
- and revenue comparison under optimal design.

It is weaker, by the authors' own admission, on environments featuring:

- strong bidder asymmetries,
- endogenous entry,
- information acquisition,
- reputational motives,
- collusion,
- term-structure issues,
- or bid restrictions.

Interpretation. The model is deliberately stylized. The paper is not claiming to solve every treasury-auction problem. It is isolating the underappreciated advantages of pay-as-bid in a benchmark environment that empirically fits some important markets quite well.

4.7 What is robust beyond the main benchmark

Two robustness messages are important:

- appendix A shows the main design insights survive when mixed strategies and elastic supply curves are allowed;

- the companion note argues that small bidder information asymmetries do not overturn the design results and yield only approximate, not drastic, deviations from the symmetric-information benchmark.

Interpretation. The paper's payoff is not a fragile theorem that disappears once one moves a step away from the benchmark. The exact formulas rely on symmetry, but the design logic is meant to be broader.

4.8 Relation to empirical patterns

The paper speaks directly to several empirical regularities:

- nearly flat bids when supply randomness is low,
- small gaps between auction prices and nearby secondary-market prices,
- rough revenue equivalence between formats in some treasury studies,
- and the tendency of smaller countries to rely more on pay-as-bid.

Interpretation. The theory's job here is not just to rank formats abstractly. It is also to explain why certain auction facts that seem puzzling from the viewpoint of uniform-price theory are much more natural in a pay-as-bid world.

5 Tables and figures

The paper's visuals are theory-building rather than regression-heavy. They walk the reader from the key boundary condition, to the shape of equilibrium bids, to comparative statics, and finally to the format comparison and the connection to empirical evidence.

5.1 Figure 1: why the minimum clearing price must equal value at the top quantity

Read:

- The figure plots a downward-sloping value curve and a lower dashed bid schedule.
- At the maximum quantity that can be won, a bidder who bids below value can profitably deviate upward a little.
- The deviation raises the quantity won discretely while increasing the payment only slightly.

A CASE FOR PAY-AS-BID AUCTIONS

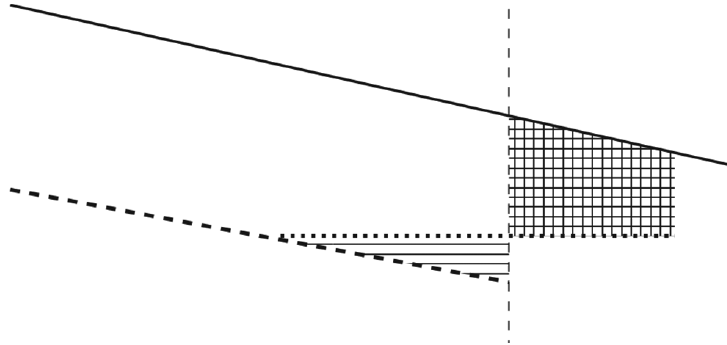


Figure 1: Paper Figure 1: intuition for theorem 1.

Economic message. The minimum clearing price is not a free object. It is pinned down by a no-profitable-deviation argument. This is the boundary condition that unlocks the rest of the equilibrium analysis.

5.2 Figure 2: equilibrium bids as weighted averages of values

Read:

- The figure shows a case with linear marginal values and truncated-normal supply.
- The bid curve lies below the value curve for most relevant quantities, illustrating bid shading.
- Bids are not flat because supply is uncertain; the weighting distribution of possible realized quantities matters.

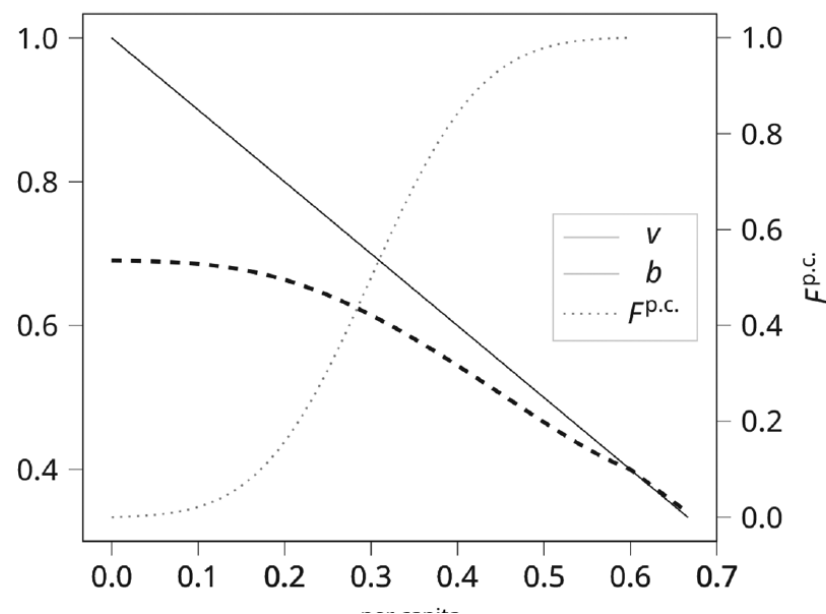


Figure 2: Paper Figure 2: nonlinear equilibrium bids when supply is uncertain.

Economic message. Pay-as-bid bids are not arbitrary step functions. They have a disciplined shape implied by theorem 3: a smooth weighted average of marginal values over higher quantities.

5.3 Figures 3 and 4: concentration of supply and number of bidders

Read:

- Figure 3 shows that when supply becomes more concentrated, bids become flatter.
- Figure 4 shows that more bidders increase bids, but the incremental effect becomes small once the market is already reasonably competitive.

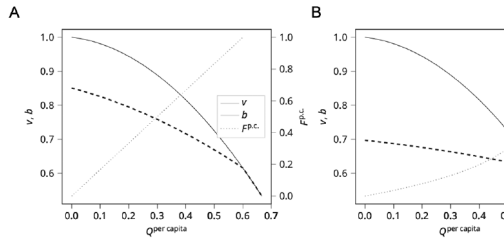


FIG. 3.—Bids (b) are flatter for more concentrated distributions ($F^{p,c}$) of supply, holding constant marginal values (v) and number of bidders ($n = 10$)

(a) Paper Figure 3

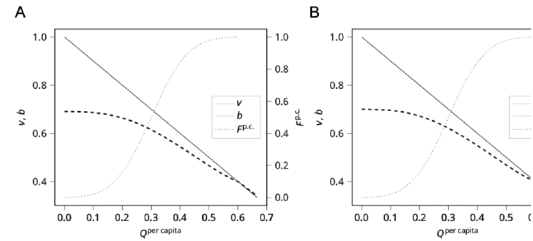


FIG. 4.—Bids for relevant quantities increase when more bidders arrive much. A, Ten bidders. B, Ten million bidders. Axis scales and per capita quantity ($F^{p,c}$; truncated Gaussian distribution with maximum per capita supply of same in both panels

(b) Paper Figure 4

Figure 3: Comparative statics for supply concentration and the number of bidders.

Economic message. These are the paper's most intuitive comparative statics. Supply certainty flattens bids, while extra competition pushes bids upward but with diminishing incremental force once the market is already fairly large.

5.4 Figure 5 and Table 1: optimal pay-as-bid versus optimal uniform-price

Read:

- Figure 5 compares values and equilibrium bids in the paper's worked example for both formats and for $n = 10$ and $n = 30$ bidders.
- In pay-as-bid, optimal reserve and per-capita quantity are essentially independent of the number of bidders in the example.
- In uniform price, both the optimal reserve and the optimal per-capita quantity move with competition because equilibrium selection and bid shading work differently.

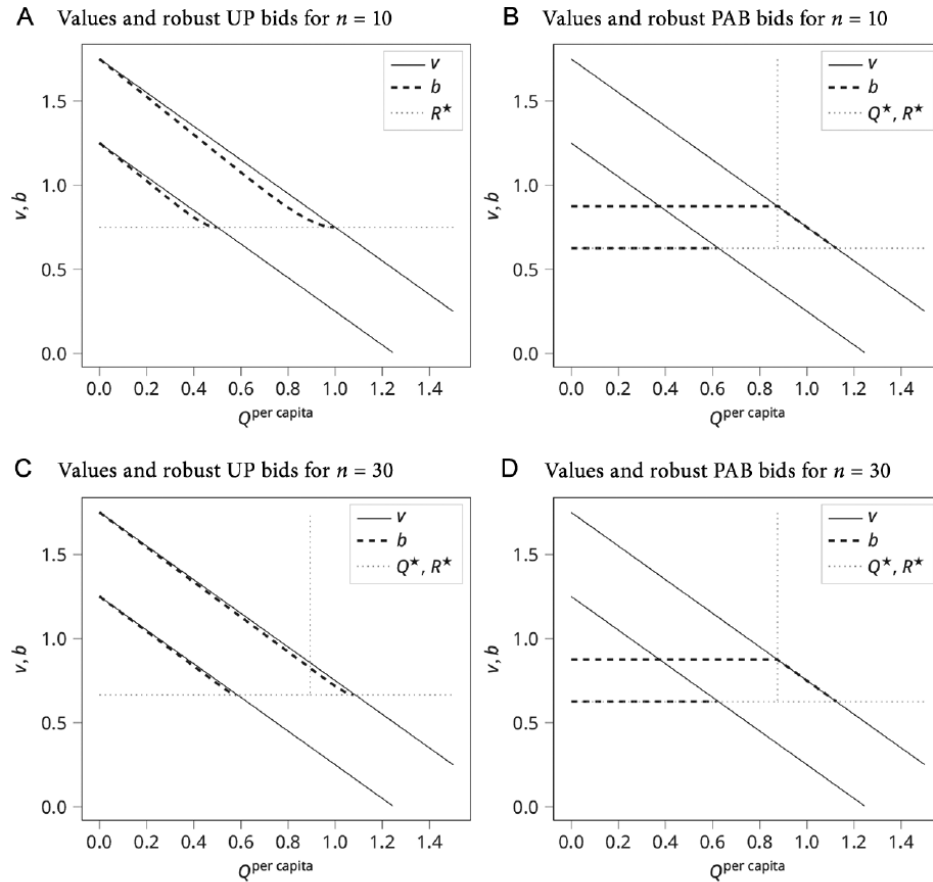


FIG. 5.—Bids in optimal pay-as-bid and uniform-price auctions. *A, B*, $n = 10$ bidders. *C, D*, $n = 30$ bidders. The horizontal dotted lines represent each auction's optimal reserve price, and the vertical dotted lines represent each auction's optimal per capita quantity. In pay-as-bid panels (*B, D*), we draw equilibrium bids that are slightly above values for a barely visible interval of quantities to the right of per capita supply. Bids beyond the optimal per capita quantity or below the optimal reserve price are irrelevant to equilibrium payoffs and are only partially determined: bids below reserve can be arbitrary, and bids beyond the maximum quantity sold can be arbitrary as long as they are sufficiently aggressive. PAB =

Figure 4: Paper Figure 5: optimal auctions under pay-as-bid and uniform price.

Table 1: Paper Table 1: expected revenue in the optimal uniform-price auction as a share of optimal pay-as-bid revenue.

Number of bidders n	10	20	30	40
$\mathbb{E}[p^{UP}]/\mathbb{E}[p^{PAB}]$ (%)	97.30	97.30	97.89	98.39

Economic message. The worked example makes the comparison concrete. Uniform price can come close, but in realistic finite markets it still leaves revenue on the table relative to an optimally

designed pay-as-bid auction. The gap shrinks with competition but does not vanish quickly in small and medium-sized markets.

5.5 Table 2: what the empirical literature looks like through the paper’s lens

Read:

- The table collects natural experiments, controlled experiments, and counterfactual exercises comparing pay-as-bid and uniform-price treasury auctions.
- In studies where noncompetitive-demand randomness is very small, pay-as-bid tends to beat or approximately match uniform price.
- In studies with much more volatile supply, the ranking is less uniform.

Table 2: Paper Table 2: revenue comparisons in government debt auctions.

Paper	Data	Method	σ/μ (%)	Bidders	Conclusion
Marszalec (2017)	Poland	PAB $\rightarrow$ counterfactual UP	0.00	12.3	PAB > UP
Barbosa et al. (2020)	China	Controlled experiment	0.00	35.2	PAB $\approx$ UP
Février, Préget, and Visser (2002)	France	PAB $\rightarrow$ counterfactual UP	1.27	20.8	PAB > UP
Armantier and Sbaï (2006)	France	PAB $\rightarrow$ counterfactual UP	3.78	19.0	UP > PAB
Hattori and Takahashi (2022)	Japan	Natural experiment	11.00	No data	PAB > UP
Umlauf (1993)	Mexico	Natural experiment	11.16	24.7	UP > PAB
Mariño and Marszalec (2020)	Philippines	Natural experiment	17.60	20.3	PAB > UP

Economic message. The paper does not claim the data always show clean dominance of one format. Instead, it uses the theory to organize the mixed evidence: supply randomness, bidder information, and equilibrium selection matter, and approximate revenue equivalence is exactly what one should expect in some environments.

6 Short summary

- The paper solves the symmetric-information pay-as-bid auction: minimum clearing price, essential uniqueness, explicit bid formula, and realistic existence conditions.
- The bid on any quantity is a weighted average of marginal values on that and higher quantities, so bids flatten as supply becomes more certain.
- The seller’s optimal pay-as-bid design is transparent: deterministic supply is better than random supply, and full disclosure of realized supply is better than opacity.

- In the seller’s design game, pay-as-bid weakly dominates uniform price in revenue. Uniform price can match pay-as-bid only in special seller-optimal equilibria.
- The theory explains why empirical revenue differences are often modest, why flat bids appear in low-uncertainty treasury auctions, and why pay-as-bid remains an underappreciated but strong practical format.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that pay-as-bid auctions are much more structured than the previous literature suggested. In the symmetric-information environment they have a unique pure-strategy equilibrium, a closed-form bid representation, and a sharply characterized design problem. That design problem has a clean answer: deterministic supply and full disclosure maximize revenue.

7.2 Economic intuition

The economic intuition is simple once theorem 1 is in place. Bidders must bid up to value at the effective maximum quantity, and the rest of the bid schedule is then a weighted averaging problem over residual supply. Because random supply and hidden supply information force bidders to hedge in ways that soften bids, the seller wants transparency. Because uniform-price auctions suffer from equilibrium multiplicity, their design problem is fundamentally less disciplined.

7.3 Takeaway

If the seller cares mainly about revenue and operates in an environment where bidders are approximately symmetrically informed, this paper says the natural benchmark is not an opaque, randomized, or equilibrium-selection-sensitive auction. The benchmark is a transparent pay-as-bid auction with deterministic quantity, a reserve price chosen like a monopolist’s price floor, and full disclosure of realized supply.